

MOTIONCORE · REQUIREMENTS

MotionCore requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/motioncore		
Source	docs/03-requirements.md @ 05f91fc		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against the concept estimates
0.1	2026-09-25	Amish Chadha	Initial scaffold

MotionCore requirements

These are first-pass requirements for the concept. Targets are proposals for review, not yet agreed with the host projects, and will be checked by calculation at TRL 3. The status column compares each target with the first-order estimates in MTC-PRC-001; no requirement has been verified by test.

The **reference case** used throughout is a 250 W geared hub motor on a 48 V (46.8 V nominal) SwapCell pack at 40 °C ambient. The **heavy case** is 500 W continuous on a 25.6 V LiFePO4 pack at 40 °C ambient.

Table 1. Requirements.

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 2
R1	Supply range	Operate from 20 to 58 V DC (8S LiFePO4 to 13S lithium-ion, including SwapCell at 54.6 V full); withstand 75 V transients from regeneration	Controller and contactor datasheets; bus voltage calculation	Met by design (estimate)
R2	Motor power	250 W continuous in the reference case; 500 W continuous in the heavy case; 750 W for 10 s	Current and thermal calculation	250 W met; 500 W at risk (see R9)
R3	Hardwired emergency stop	Twin-channel normally closed e-stop wired in series with the contactor coil; motor power removed within 100 ms of actuation; works with the supervisor and controller both failed	Circuit review; contactor drop-out time from datasheet	Met on paper (about 60 ms estimated); unverified
R4	No automatic restart	After an e-stop, power loss or fault, the drive stays off until the e-stop is released, the throttle is at zero and the operator cycles the enable key	Circuit and state-machine review	Met by design; unverified
R5	Brake interlock	Two brake switch inputs; drive torque removed within 200 ms of either switch opening; a spring-applied brake output released only while the safety loop is closed	Circuit review; supervisor timing budget	Met on paper; unverified
R6	Independent speed limit	Speed limit set in the controller and checked by the supervisor from a separate sensor; overspeed of more than 10 % for more than 0.5 s opens the contactor; defaults 25 km/h (pedal assist) and 1.5 m/s (walk-behind)	Firmware review; sensor resolution calculation	Met on paper; unverified
R7	Fault reaction	Supervisor removes torque within 200 ms of: command timeout, throttle signal outside 0.5 to 4.5 V, loss of controller CAN heartbeat, over-temperature or BMS fault flag	Fault table review	Met on paper; unverified
R8	Precharge	Inrush at contactor closure under 5 A; bus ready within 1 s	Precharge calculation	Met (about 0.55 A peak, 0.5 s, estimate)
R9	Thermal	Enclosure surface 60 °C or less at 40 °C ambient in natural convection	Thermal calculation	Reference case met (about 55 °C, estimate); heavy case not met (about 67 °C, estimate)
R10	Ingress and vibration	Enclosure and mated connectors IP65; survive road and stair vibration without loosening	Datasheets and design review	Not yet assessed
R11	One interface	One documented connector set (power in, motor, sensors, safety loop, command) keyed against mis-mating; fitted to a new host in 2 h or less with hand tools	Interface document; timed fit at TRL 4	Interface drafted in MTC-PRC-001; unverified
R12	Size and mass	Module within 250 x 170 x 70 mm; module mass 1.5 kg or less	Model and mass estimate	Met (about 250 x 170 x 66 mm, about 1.3 kg, estimate)

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 2
R13	Open and repairable	Hardware under CERN-OHL-S-2.0; supervisor firmware MIT; controller firmware open source; fault log readable over USB without closed software	License review	At risk: VESC firmware is GPL-3.0, so its relation to the MIT supervisor firmware needs review
R14	Cost	Reference kit, including the 250 W motor, \$300 or less in parts	Priced BOM (bom/ bom.csv)	Not met: about \$325 (indicative)

■ Assumptions

- Motor efficiency about 80 % and controller efficiency about 95 % at 250 W, typical of small geared hub motors and VESC-class controllers; to be checked against datasheets at TRL 3.
- Contactor drop-out time 20 to 50 ms with a Zener coil suppressor, typical of small DC contactors; to be confirmed from the chosen part.
- Enclosure heat transfer about 10 W/m²K (natural convection plus radiation) over about 0.13 m² including fins.
- The host provides mechanical brakes able to stop it with the motor unpowered. MotionCore removes drive power; it does not replace the host's brakes.
- The host carries out its own risk assessment. MotionCore targets are not a claimed ISO 13849 performance level.